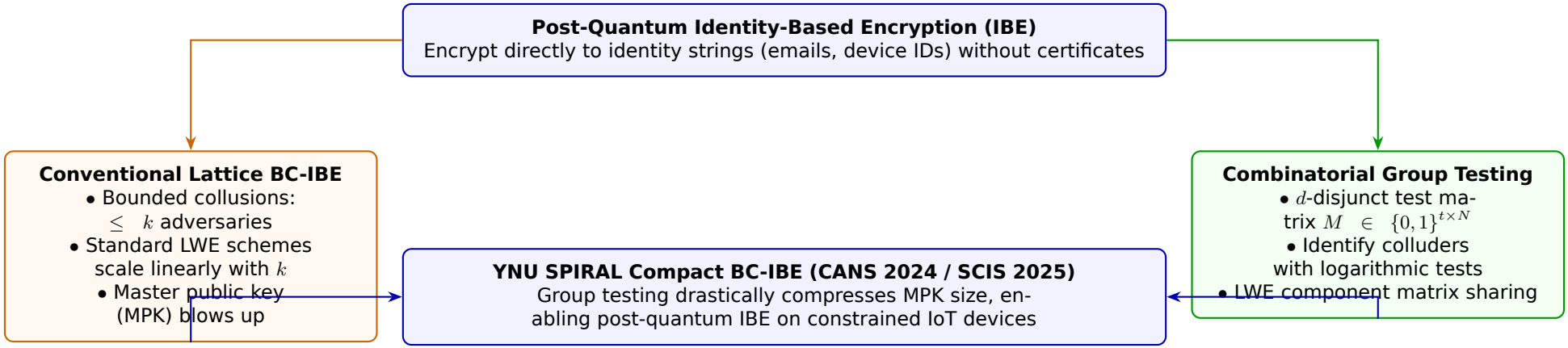

Post-Quantum Identity-Based Encryption (IBE)
Encrypt directly to identity strings (emails, device IDs) without certificates
Conventional Lattice BC-IBE
• Bounded collusions: $\leq k$ adversaries
• Standard LWE schemes scale linearly with k
• Master public key (MPK) blows up
Combinatorial Group Testing
• d-disjunct test matrix $M \in \{0,1\}^{t \times N}$
• Identify colluders with logarithmic tests
• LWE component matrix sharing
YNU SPIRAL Compact BC-IBE (CANS 2024 / SCIS 2025)
Group testing drastically compresses MPK size, enabling post-quantum IBE on constrained IoT devices